CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**GEO SYNTHETICS AND REINFORCED EARTH STRUCTURES** **Professional Elective-IV**	L	T	P	C
23A01702a		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the concept and applications of reinforced earth, including friction coefficient determination.
2. Analyze the classification, functions, and durability aspects ofgeosynthetics and their advantages over conventional materials.
3. Design reinforced earth retaining walls considering stability mechanisms and material selection.
4. Evaluate the performance of reinforced embankments and their foundation mattresses for settlement and stability control.
5. Develop reinforced soil beds and analyze reinforced pavements using standard design approaches.

Course Outcomes (COs):

After successful completion of this course, students will be able to:

1. Explain the fundamentals of reinforced earth and analyze the effects of reinforcement on soil properties.
2. Compare different types ofgeosynthetics, their functions, and durability aspectsin geotechnical applications.
3. Design reinforced earth retaining walls considering stability mechanisms and reinforcement layouts.
4. Evaluate reinforced embankments and foundation mattresses with respect to settlement and load-bearing capacity.
5. Design and analyze reinforced pavements and soil beds using standard methodologies.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	2	1	1	1	-	1	1	-	-	1	-	1	1	1
CO -2	2	1	2	1	1	-	1	-	-	-	-	1	1	1
CO -3	2	1	2	1	1	-	1	-	-	-	-	1	1	1
CO -4	2	1	2	1	1	1	1	-	-	-	-	1	1	1
CO -5	2	1	2	1	1	1	2	-	-	-	-	1	2	1

UNIT – I

Reinforced Earth: Concept, Effects of Reinforcement On Soils – Equal Confining andPsuedo Cohesion Concepts, Materials, Friction Coefficient – Definition, Laboratory Determination, Factors Affecting Fiction Coefficient; Application of Reinforced Earth

UNIT – II

Geosynthetics - Advantages Over Conventional Materials - Classification Based On Material Type and Function - Types of Geosynthetics - Functions of Geosynthetics - Tests On

Geosynthetics - Durability Aspects of Geosynthetics - Applications of Geosynthetics

UNIT – III

Reinforced Earth Retaining Walls: Introduction, Stability Mechanisms, Design of Reinforced Earth Retaining Wall - Selection of Materials - Geotechnical Analysis - Reinforcement Layout and Spacing - Stability Analysis - Advantages Over Conventional Retaining Walls

UNIT – IV

Reinforced Embankments: Introduction, Design of Reinforced Embankment, Foundation Mattress Below The Embankment - Purpose and Function of Foundation Mattresses - Components of Reinforced Mattress - Design of Reinforced Mattress - Design Calculations for Settlement Control, Bearing Capacity, and Long-Term Performance. Field Implementation and Monitoring Techniques

UNIT – V

Reinforced Soil Beds: Introduction, Factors Affecting The Behaviour of Reinforced Soil Beds, Analysis and Design Reinforced Pavements: Benefits of Placing Reinforcementin Flexible Pavement Layers, Design of Reinforced Pavements By Giroud andNoiray Approach and Modified CBR Method.

TEXT BOOKS:

1. An Introduction to Soil Reinforcement and Geosynthetics" By G.L. Siva Kumar Babu, University Press
2. Fundamentals of Geosynthetics Engineering, Sanjay Kumar Shukla andJian-Hua Yin, CRC Press, 2017, 1st edition.
3. Reinforced Soil and its Engineering Applications, Swami Saran, I.K. International Publishing House Pvt. Ltd., 2019, 1st edition.

REFRENCE BOOKS:

1. Designing with Geosynthetics by Robert M Koerner, R.M. Pearson Education Inc., 2012, 6th edition
2. Advances in Geosynthetics by G. Venkatapparao, Sai Master Geo environmental Services Pvt. Ltd. Publications
3. Designing with Geosynthetics, Koerner, R.M., Pearson Education Inc., 2012, 6th edition
4. IS:13162-1992; IS:14293& 94-1995; IS:14324-1995; IS:14714-1999, Geotextiles – Methods of Tests
5. IRC: SP:102-2014: Guidelines for design and construction of reinforced soil walls

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/106/105106052/
https://onlinecourses.nptel.ac.in/noc20_ce06/preview